

foamin: A Deformable Sensor for Multimodal Inputs Based on Conductive Foam with a Single Wire

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ABSTRACT

Soft sensors made of deformable materials, that are capable of sensing touches or gestures, have attracted considerable attention for use in tangible interfaces or soft robotics. However, to achieve multimodal gesture detection with soft sensors, prior studies have combined multiple sensors or utilized complex configurations with multiple wires. To achieve multimodal gesture sensing with a simpler configuration, a novel soft sensor consisting of a conductive foam with a single wire, was proposed in this study. This sensor was named as foamin and utilizes an impedance measurement technique at multiple frequencies called foamin. Additionally, a surface-shielding method was designed for improving the detection performance of the sensor. Several patterns of foamin were implemented to investigate the detection accuracy and three application scenarios based on the sensor were proposed.

CCS CONCEPTS

• Human-centered computing → Interaction devices.

KEYWORDS

Deformable Interfaces, Soft Material, Touch Sensing

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1 INTRODUCTION

In recent times, most of the computer interfaces used are composed of hard materials and parts, however studies on interfaces composed of soft materials and non-rigid interactions has been attracting attention. Although several types of soft sensors have been proposed, they still have a limitation where the number of input modes that can be identified by a single sensor is not large compared with the deformation capability of the material. Using a soft sensor to detect multiple deformation modes typically requires the embedding of a hard sensor in a soft material or the use of multiple wires, that increases the cost, size, and complexity of the

system. It is therefore, useful to detect multiple gestures with a system consisting of a soft sensor with less wire.

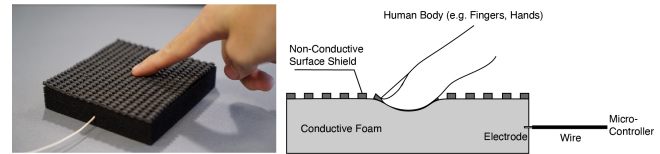


Figure 1: foamin.

In this study, we propose a novel sensing method for detecting deformable gestures, such as touching, compressing, pinching, and grasping by the human body through a single wire connected to a soft object made of a single foam material (see Fig. 2.) For touch sensing, swept-frequency capacitive sensing (SFCS) is often used to detect various touch gestures by measuring the impedance of multiple frequencies with a single wire. While this method has been applied to hard conductive materials with less deformation, one of the originalities of our study is that we applied this method for deformation sensing of soft materials as well as touch sensing. The foam-based multimodal soft sensor is named as “foamin” in this study. The contributions of this study are as follows:

- proposed a soft sensor connected to a single wire based on SFCS to identify multiple touch and deformation inputs,
- a surface design with a mesh shield to increase detection performance,
- investigated detection accuracy and distance resolution,
- proposed application examples based on the foamin sensor,

Below, the mechanism of the foamin is described, the sensing performance of implemented foamin is reported, and application examples to demonstrate the potential of foamin as a new interface is proposed.

2 RELATED WORK

Deformation sensors made of soft materials have been proposed in the field of HCI and robotics based on the trend of softening the interface of products interacting with human beings. For deformation sensing, some studies have combined soft materials and hard sensors. Flexibles [6] detected the deformation of 3D soft objects using a display with a capacitive sensor. Fuwafuwa [7] sensed its deformation using embedded optical sensors.

As another approach, some researchers have attempted to detect deformities by attaching multiple electrodes to soft objects. iSoft [8] used elastomers for the sensor and detected touch and stretch using electrical impedance tomography (EIT). FoamSense [3] used a porous material with conductive ink to detect the deformation. It

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detects deformations, such as compression, bending, torsion, and shearing by impregnating a porous material with conductive ink. However, these sensors require complex cabling to detect multi-modal deformation inputs.

Physical interfaces for hand and finger gesture detection have also been extensively studied for HCI. Tomo [9] has detected finger touch gestures on skin using EIT. Electrck [10] combined commercially available conductive materials and electric field tomography to provide low-cost touch detection. ScraTouch [2] achieved touch sensing using fingernails by detecting the change in leakage current to the human body owing to the change in impedance. Among these pioneering studies, we gained inspiration from studies using SFCS for touch detection [4, 5]. This study contributes to the sensor design, therefore SFCS can be applied for deformation and touch gesture sensing of soft objects.

3 SYSTEM DESIGN OF FOAMIN

Our proposed sensor, foam-in, consists of a soft conductive foam, a shield that covers the surface, and a single cable that connects to the microcontroller. The sensor is entirely made of soft material, and it can acquire and identify the user's actions of touching and deforming it through the fingers and other body parts.

3.1 Conductive Foam and Circuit

Fig. 2 shows the configuration of the foam-in sensor. In this system, we measured the impedance of the system at multiple frequencies through a sensor board based on the SFCS method.

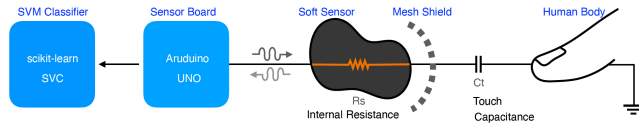


Figure 2: System overview of foam-in.

Although various types of porous materials could be used for foam-in as FoamSense [3], a conductive polyurethane foam was chosen as the body of the soft sensor. To model the configuration in Fig. 2 (a) as an electric circuit, the internal resistance R_s of the soft sensor and the touch capacitance C_t between the person and the soft sensor can be regarded as an RC series circuit. In this case, the impedance Z can be expressed as Equation (1).

$$Z = \sqrt{R_s^2 + \frac{1}{(\omega C_t)^2}} \quad (1)$$

Similar to the Touché sensor, the configuration of the foam-in can also recognize touch input on its surface by fingers or hands. Furthermore, when the user inputs deform the sensor body part, the impedance Z changes significantly not only with the change of contact area size because of the touch but also with the structural change owing to the compression of the body and the distance between the wire contact and the touch point. By acquiring the impedance changes associated with these factors, the different modes and positions of input could be identified.

3.2 Surface Shield

When a soft material is used against the SFCS, the contact area changes according to the different angles of the fingers and the amount of push-in, although the same finger touches it, making it difficult to distinguish and detect impedance changes owing to material deformation or position. To constrain the change in the contact area, we designed a surface shield that covers the conductive foam with a mesh-patterned insulating sheet, as shown in Fig. 2 (b). This makes it easier to distinguish between touch and compression gestures by ensuring that the conductive foam is touched directly only when a force greater than a certain threshold is applied.

3.3 Classification

In this study, an Arduino UNO was used and a program was developed to run the SFCS based on Equation [1]. In the SFCS, the AC voltage sweeps at 184 different frequencies in the range of 35 kHz to 200 kHz. The swept frequencies were selected in the interval of 2 kHz to 5 kHz. As the classifier, a linear support vector machine (SVM) was adopted on the scikit-learn. First, we preprocessed the raw signal outputs responding to 184 PWM wave inputs with time-averaging, simple moving averaging of five adjacent frequencies, and standardization of signal amplitude on the interval [0,1], which are then used as 180 features in SVM.

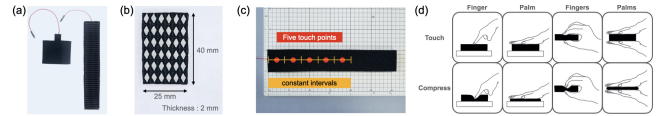


Figure 3: (a) Testing devices. (b) Surface shield used for testing. (c) Measurement method. (d) Deformable gesture set.

4 IMPLEMENTATION AND SENSING PERFORMANCES

Based on the above design, we implemented two types of foam-in sensor, as shown in Fig. 3 (a). The conductive foam that was used had a volume resistivity of less than $10^5 \Omega \text{ cm}$, a hardness of less than 5 Asker C, and a density of 0.03 g/cm^3 . The surface shield that was used had a mesh-like design, as shown in Fig. 3 (b). It had a thickness of 2 mm and reduced the contact area by one-third when being pressed.

Two types of tests were conducted to investigate the sensing accuracy of foam-in. For these tests, 10 types of input gestures were determined to be identified on the foam-in. First, the profile data of each gesture input was collected to be used for the SVM in advance.

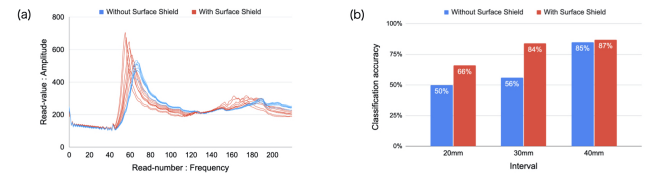


Figure 4: (a) Impedance profile when detecting the eight gestures. (b) Accuracy of detecting distance and gestures

For this, the data was obtained by holding each gesture for 4 seconds. Because the program worked at 5 fps and, the process was repeated for three times. In total 60 sets of data per gesture was collected as training data.

After the training process, the test data obtained with the foamin was classified using SVM and its accuracy was tested. Because the classification results could be unstable during the deformation, the data after the deformation stabilized was used as the test data. Specifically, the data was obtained when the classification result had not been changed for more than one second.

Based on this protocol, first the gesture classification accuracy was investigated with a rectangular foamin sensor (100 mm × 80 mm × 20 mm), as shown in Fig. 3 (a) left. Then the classification of eight different types of gesture inputs were tested, as shown in Fig. 3 (d).

Each gesture input was conducted for 10 times (in total 80 times). When the foamin was used without the mesh shield, it successfully classified the gestures with an accuracy of 81%. In contrast, when the mesh shield was attached to the surface, the test gestures were classified with an accuracy of 100%. Fig. 4 (a) shows the average impedance profile used for the training data. When the surface shield is applied to the sensor, it can be seen that the impedance profile changes significantly for each gesture.

Next, we tested how accurately foamin can estimate the distance between the contact point of the wire and the point where the gesture was made. Using a rectangular conductive foam of 50 mm × 300 mm × 20 mm, the classification of two different types of gestures were tested (touch and compression with a single finger) at five points. Fig. 3 (c) shows an example of five points and this test clarifies the distance resolution of foamin. Fig. 4 (b) shows that the distance resolution was improved by the addition of a surface shield. Without the surface shield, the gesture classification accuracies were 50%, 58%, and 85% when the interval distances were 20, 30, and 40mm, respectively. On the contrary, with the surface shield, the accuracies were 66%, 84%, and 87% in each interval condition.

5 APPLICATION EXAMPLES

Foamin can be implemented in various situations since it allows to use a conductive foam of different sizes and shapes just connected with a single wire. We implemented three applications that demonstrated various capabilities.

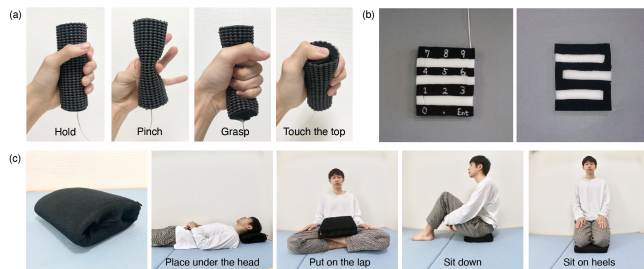


Figure 5: (a) Deformable musical instruments, (b) Soft numeric keypad, (c) Activity sensible cushion

5.1 Deformable Musical Instruments

Using the foamin, we implemented a musical instrument that changes its sound depending on deformable gestures. This instrument detects the four gestures shown as Fig. 5 (a) and makes different sounds when touching the top, holding it, pinching it, and grasping it.

5.2 Soft Numeric Keypad

Based on the touching point estimation capability, we implemented a numeric keypad with 12 buttons. This keypad can detect a user's pressing actions using a single finger. The keyboard was made of black conductive foam and white non-conductive foam bonded together with double-sided tape. As shown in Fig. 5 (b), the bottom of the keyboard is a single conductive foam with a zigzag shape.

5.3 Activity Sensible Cushion

Taking advantage of the foamin's scalability, we implemented a rounded pillow with a size of 250 mm × 250 mm × 100 mm. This cushion is made by layering 20 mm-thick conductive foams and covering it with a mesh shield. In the current implementation, this cushion can identify different postures, as shown in Fig. 5 (c). These inputs are expected to be used in applications, such as health monitoring.

6 CONCLUSION AND FUTURE WORK

In this paper, we proposed a novel deformable sensor named foamin, that is capable of detecting multiple touch or deformation inputs with machine learning. Unlike conventional soft sensors, foamin can be easily created by simply connecting a single wire to a conductive foam, and therefore has a wide range of applications.

In future work, we need to find an appropriate wiring design for each application. Additionally, we will explore further ways to detect and identify complex deformation modes such as twist, shear, and bending, based on the foamin method. Finally, we will further develop a further design of a mesh shield attached to the foam surface and clarify the design space of this sensor more.

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